

ARCHIE BENN

HOW TO TRAIN YOUR MODEL

ASSESSING THE IMPACTS OF FOREST AGE ON THE
SPATIAL AND TEMPORAL GENERALISATION OF RANDOM
FOREST EVAPOTRANSPIRATION MODELS

MSc Bioinformatics

Supervisor: Martin De Kauwe

May – September 2026



University of
BRISTOL

A dissertation submitted to the University of Bristol
in accordance with the requirements of the degree of
Master of Science by advanced study in Bioinformatics
in the Faculty of Health and Life Sciences.

ABSTRACT

Nulla ac nisl. Nullam urna nulla, ullamcorper in, interdum sit amet, gravida ut, risus. Aenean ac enim. In luctus. Phasellus eu quam vitae turpis viverra pellentesque. Duis feugiat felis ut enim. Phasellus pharetra, sem id porttitor sodales, magna nunc aliquet nibh, nec blandit nisl mauris at pede. Suspendisse risus risus, lobortis eget, semper at, imperdiet sit amet, quam. Quisque scelerisque dapibus nibh. Nam enim. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nunc ut metus. Ut metus justo, auctor at, ultrices eu, sagittis ut, purus. Aliquam aliquam.

Key words: Machine Learning, Evapotranspiration, Forest Age, FLUXNET.

Word count: 4000

DEDICATION AND ACKNOWLEDGMENTS

DECLARATION

I declare that the work in this dissertation was carried out in accordance with the requirements of the University's Regulations and Code of Practice for Taught Programmes and that it has not been submitted for any other academic award. Except where indicated by specific reference in the text, this work is my own work. Work done in collaboration with, or with the assistance of others, is indicated as such. I have identified all material in this dissertation which is not my own work through appropriate referencing and acknowledgment. Where I have quoted or otherwise incorporated material which is the work of others, I have included the source in the references. Any views expressed in the dissertation, other than referenced material, are those of the author.

Archie Benn

August 17, 2026

CONTENTS

1	Aims and Objectives	5
2	Methods	5
2.1	Data Acquisition and Setup	5
2.2	Cluster Analysis	5
2.3	Random Forest modelling	6
2.4	A Two Site Case Study	7
3	Results	7
3.1	Site Clustering	7
3.2	Spatial Generalisation	10
3.3	Temporal Generalisation	12
3.4	Case Study of US-UMd and DE-Lnf	12
4	Discussion	14
5	Conclusion	14
6	Figures and Tables	17
7	Supplementary Materials	17

1 AIMS AND OBJECTIVES

2 METHODS

2.1 Data Acquisition and Setup

Flux tower data were downloaded from the FLUXNET2015 dataset [1] and processed using R (version 4.5.3) [2] with the tidyverse [3] suite of packages. Flux data were filtered to only keep North American and European sites whose forest ages are dated in Besnard et al. [4], with rows of variables within these sites filtered based on their provided FLUXNET QC flag value. Four-day LAI data (NASA product: MCD15A3H [5]) were retrieved using site coordinates via MODISTools [6], filtered on QC, and linearly interpolated between the four-day periods of LAI to provide a continuous daily value, before joining by date and site to the main FLUXNET dataset.

Site stand ages retrieved from Besnard et al. [4] are static despite multi-year data ranges often contained at these sites in the FLUXNET. As such, site ages were extrapolated across the range of data at each site to act as a continuous metric, with the provided age set to the first row of the FLUXNET data for each site. Köppen climate sites were retrieved and attached based on site latitude and longitude, and these classifications were further simplified to their main groups (Temperate and Continental in this case). A two week cumulative sum of precipitation prior to the day of measurement was also calculated and attached per row, with the first 13 days of data at each site then being discarded.

2.2 Cluster Analysis

Cluster analysis is an exploratory technique which aims to classify objects into as-yet-unknown groups based on their similarities [7]. To group sites in this study using cluster analysis, a Gower's distance matrix was first formed. Gower's distances are a similarity measure which can handle both numerical and categorical data to provide a distance value between two data points (sites) from 0 (completely dissimilar) to 1 (completely similar) [8]. A table of the site features used for these distance calculations can be seen in Table 1. The Partitioning Around Medoids (PAM) algorithm and silhouette method was then used to map the Gower's distance matrix into a specified set of clusters [9]. PAM identifies medoids, which are the points within each cluster whose sum of dissimilarities to all other sites in the same cluster is minimal, and so, in this case, can be identified as the most representative flux tower sites of each cluster.

Table 1: Variables and definitions used to calculate Gower's distances during cluster analysis.

Variable	Data Type	Definition
ET	Numerical	Mean daily ET rate (mm d^{-1})
T _{air}	Numerical	Mean air temperature at site ($^{\circ}\text{C}$)
SW _{rad}	Numerical	Mean daily incoming short-wave radiation (W m^{-2})
VPD	Numerical	Mean daily vapour pressure deficit (hPa)
Site _{age}	Numerical	Mean forest stand age at site (years)
Wspeed	Numerical	Mean daily wind speed (m s^{-1})
P _{sum_14D}	Numerical	Mean two-week cumulative sum of precipitation prior to the day of measurement (mm)
Lai _{500m}	Numerical	Mean daily leaf area index ($\text{m}^2 \text{m}^{-2}$)
Cover _{type}	Categorical	IGBP cover type classification of site
Climate _{zone}	Categorical	Main Köppen climate group of site

2.3 Random Forest modelling

In this study, Random Forest models were trained and tested with `RandomForestRegressor()` from `scikit-learn` [10] using the different methods outlined below. Due to time constraints, a single set of optimised parameter values was calculated once using `GroupKFold()` and `Optuna` [11] to minimise root mean square error (RMSE) on fold predictions. These values were used for all RF models throughout the study (see Table 2). In all cases, metrics for R^2 , RMSE, and mean absolute error (MAE) were calculated using their respective `scikit-learn` functions.

Table 2: Parameters selected for use in all RF training setups in this study. Calculated to minimise RMSE on trials using `GroupKFold()` and `Optuna`.

RF parameter	Value
n_estimators	450
max_depth	20
min_samples_leaf	5
max_features	0.6

2.3.1 Model Sensitivity Testing

To assess the sensitivity of the RF models to different features, a forward feature selection assessment was carried out. Non-target (ET) numerical features were added sequentially and the RF model was re-trained and tested with each new feature added. Features were added in the following order: T_{air}, SW_{rad}, VPD, Wspeed, P_{sum_14D}, Lai_{500m}, Site_{Age}. Training sites consisted of all sites apart from medoids (see ??), with predictions formed and testing carried out at medoid sites. Results can be found in section 3.2.1.

2.3.2 Random Sampling of Training Data

To investigate how predictive performance is impacted by including site age as a feature, two separate RF architectures were formed: ‘Mod’ (Equation 1) and ‘Mod + Age’ (Equation 2).

$$ET_i = RF_{Mod}(T_air_i, SW_rad_i, VPD_i, W_speed_i, \sum_{i=13}^i P_i, LAI_i) \quad (1)$$

$$ET_i = RF_{Mod+Age}(T_air_i, SW_rad_i, VPD_i, W_speed_i, \sum_{i=13}^i P_i, LAI_i, Site_Age_i) \quad (2)$$

where the target ET_i is the estimated ET rate on the i th day (mm/day); $RF(\cdot)$ is the Random Forest model; the features $T_air, SW_rad, VPD, W_speed, \sum_{i=13}^i P_i, LAI, Site_Age$ are all listed with descriptions in Table 3.

2.3.3 Time Series Decomposition

2.4 A Two Site Case Study

3 RESULTS

3.1 Site Clustering

Cluster analysis from Gower distances and

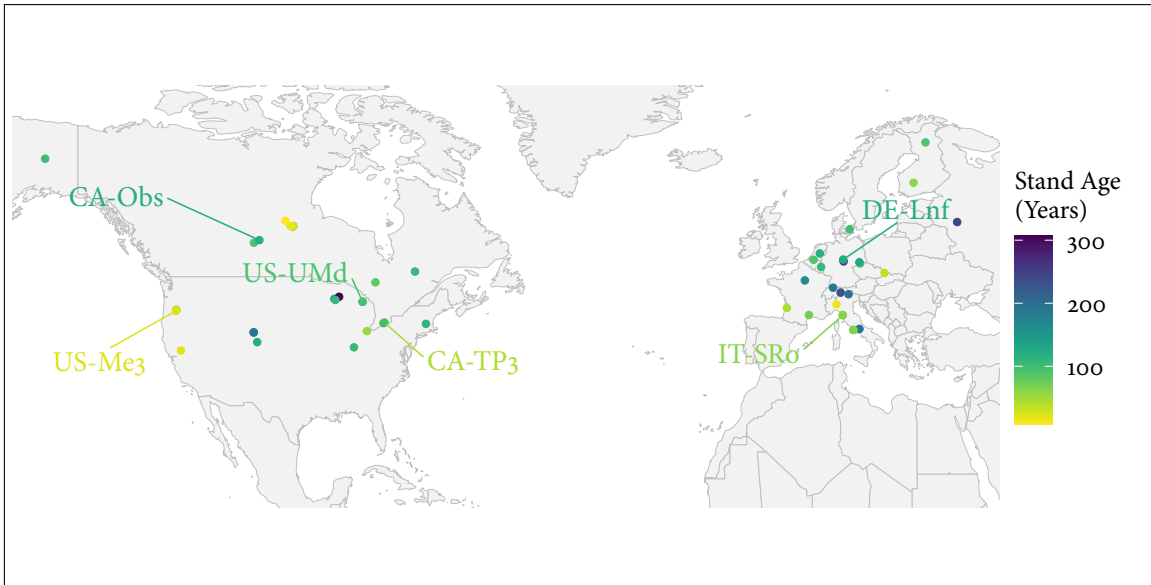


Figure 1: Map of FLUXNET sites and associated forest stand age in this study, with medoid cluster site IDs highlighted.

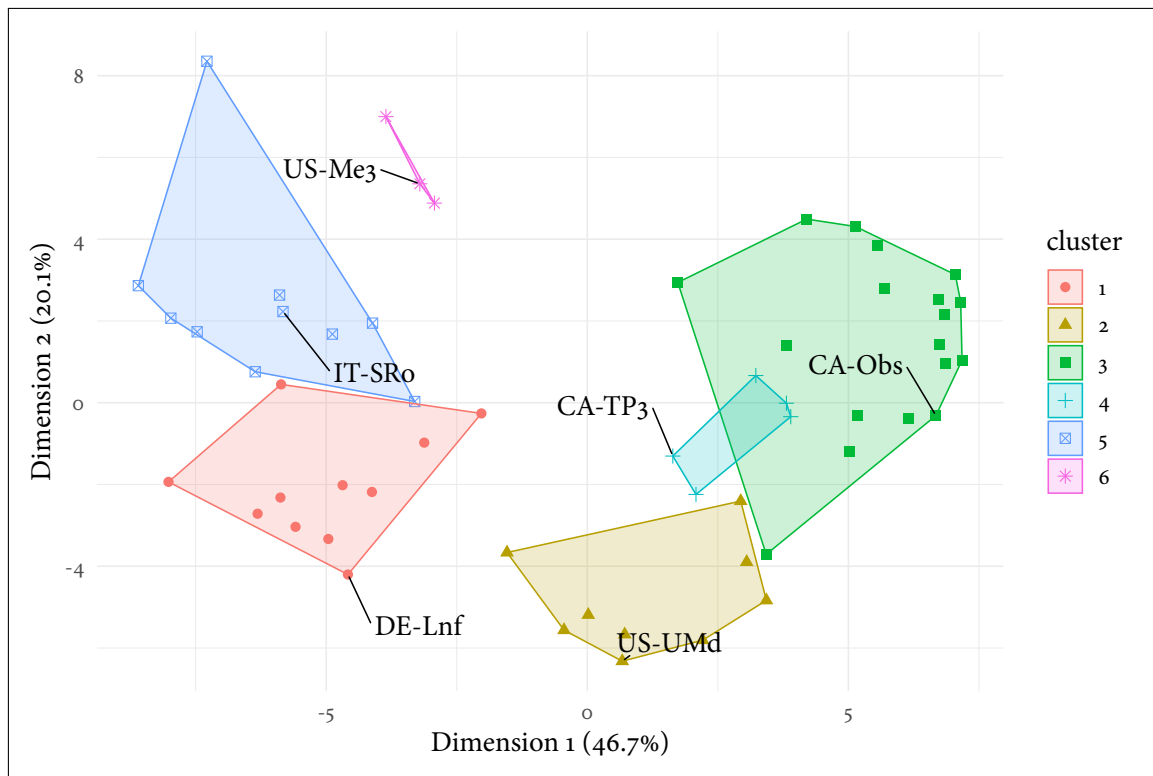


Figure 2: Cluster plot indicating the site positions in the space of the first two principal components, with coloured clusters indicating the PAM solution.

Table 3: Main characteristics of each cluster defined during cluster analysis. To see all clusters on a map see Fig. X (supplementary materials).

Cluster	Medoid Site	Defining Characteristics
1	DE-Lnf	Temperate sites without significant features.
2	US-UMd	Continental sites without significant features.
3	CA-Obs	Continental sites with low mean air temperatures.
4	CA-TP3	Continental sites with high precipitation.
5	IT-SRo	Temperate sites with high air temperatures.
6	US-Me3	Temperate sites with very high VPD, high SW radiation, low stand age, low wind speeds, and low precipitation.

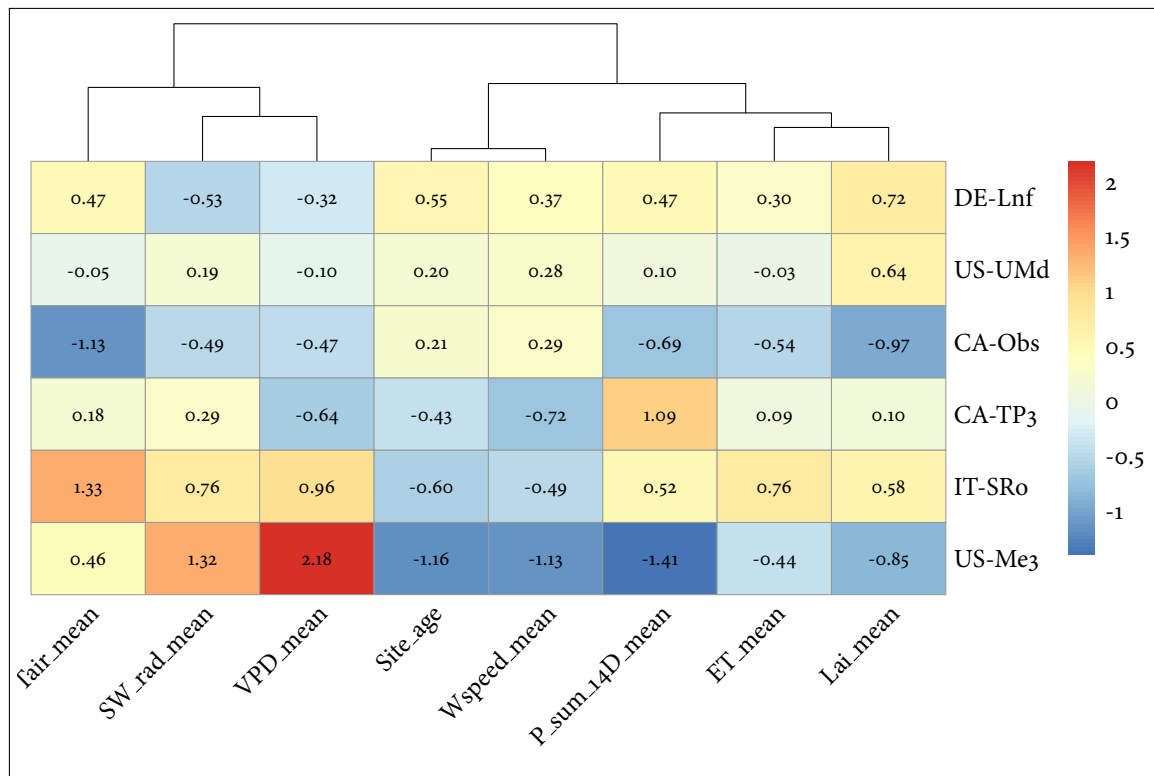


Figure 3: Heatmap of cluster medoid sites and scaled feature values. Scaled features ≥ 1.5 standard deviations from the mean of all sites were classed as 'very' (high or low, depending on direction), while features between 1 and 1.5 standard deviations from the mean were classed as high or low, depending on direction, in order to characterise clusters.

3.2 Spatial Generalisation

3.2.1 Analysis 1.1 - Model Sensitivity

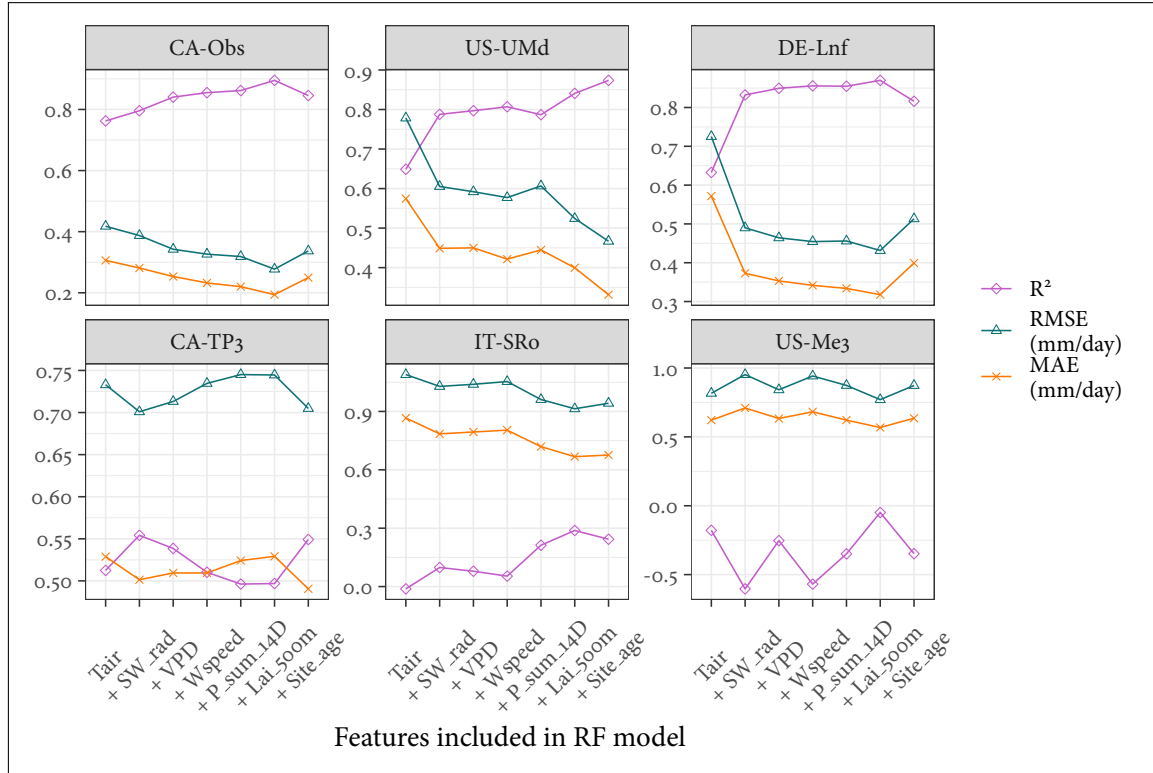


Figure 4: R², RMSE, and MAE for medoid cluster site predictions of ET (ordered left-right and top-bottom) by peak R² with clusters indicated in facet names. Models built and tested using a consecutively greater number of input features from FLUXNET and LAI data (the '+VPD model' incorporates Tair, SW_rad, and VPD as training and input features here). R² performance at only half of the sites was good, and also suggests that adding features up to site age improved prediction accuracy at four of the six medoid sites. Adding age as a final feature led to mixed performance across the medoid sites.

3.2.2 Analysis 1.2 - Shuffled Training Sets

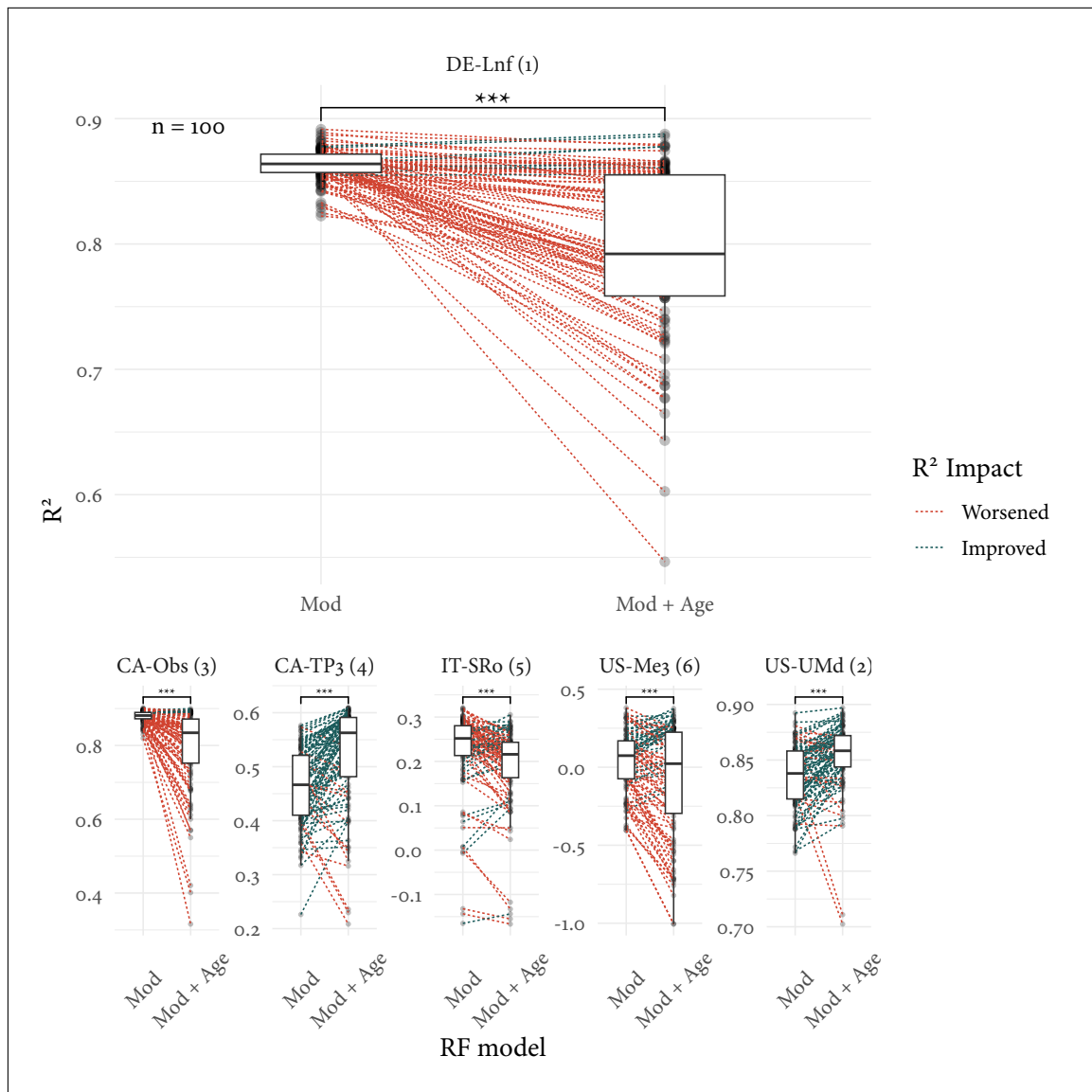


Figure 5: Paired box-plots indicate the impact on R^2 when age is added as a feature to RF models across 100 repeated runs of randomly sampled training datasets. Points are connected on a per-run basis where training dataset site composition is identical. Wilcoxon signed-rank tests suggest model performances differ significantly at all test sites. DE-Lnf is shown large for readability, with other cluster medoid sites shown at a reduced scale for comparison.

TRAINING SET COMPOSITIONS Here I will quickly look into the training set compositions of the RF models where age improved OR worsened R^2 .

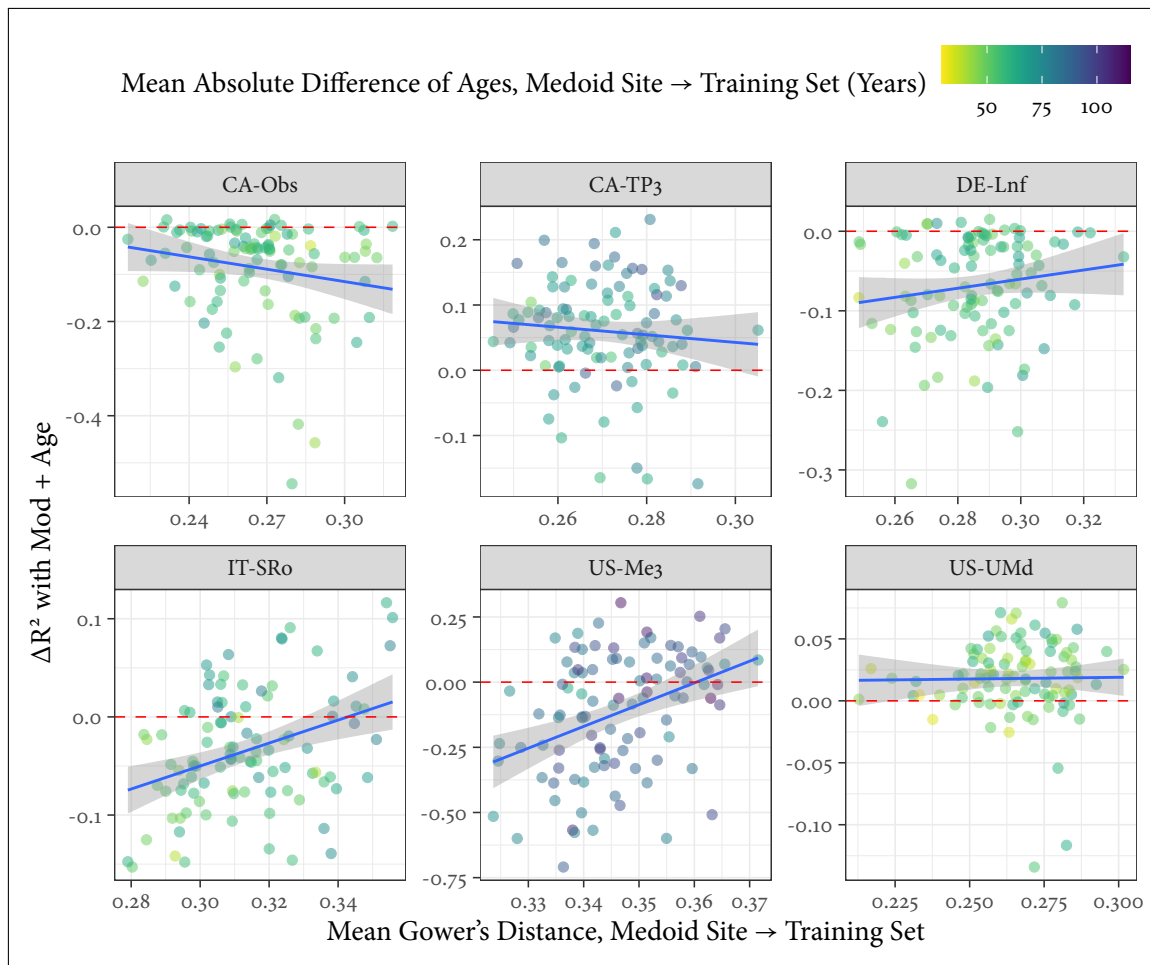


Figure 6: Plots

3.3 Temporal Generalisation

3.4 Case Study of US-UMd and DE-Lnf

- Look into predictions vs observed at both this R² plot but also over timeframe
- US-UMd: Continental Site, 2715 days of data
- DE-Lnf: Temperate Site, 2670 days of data

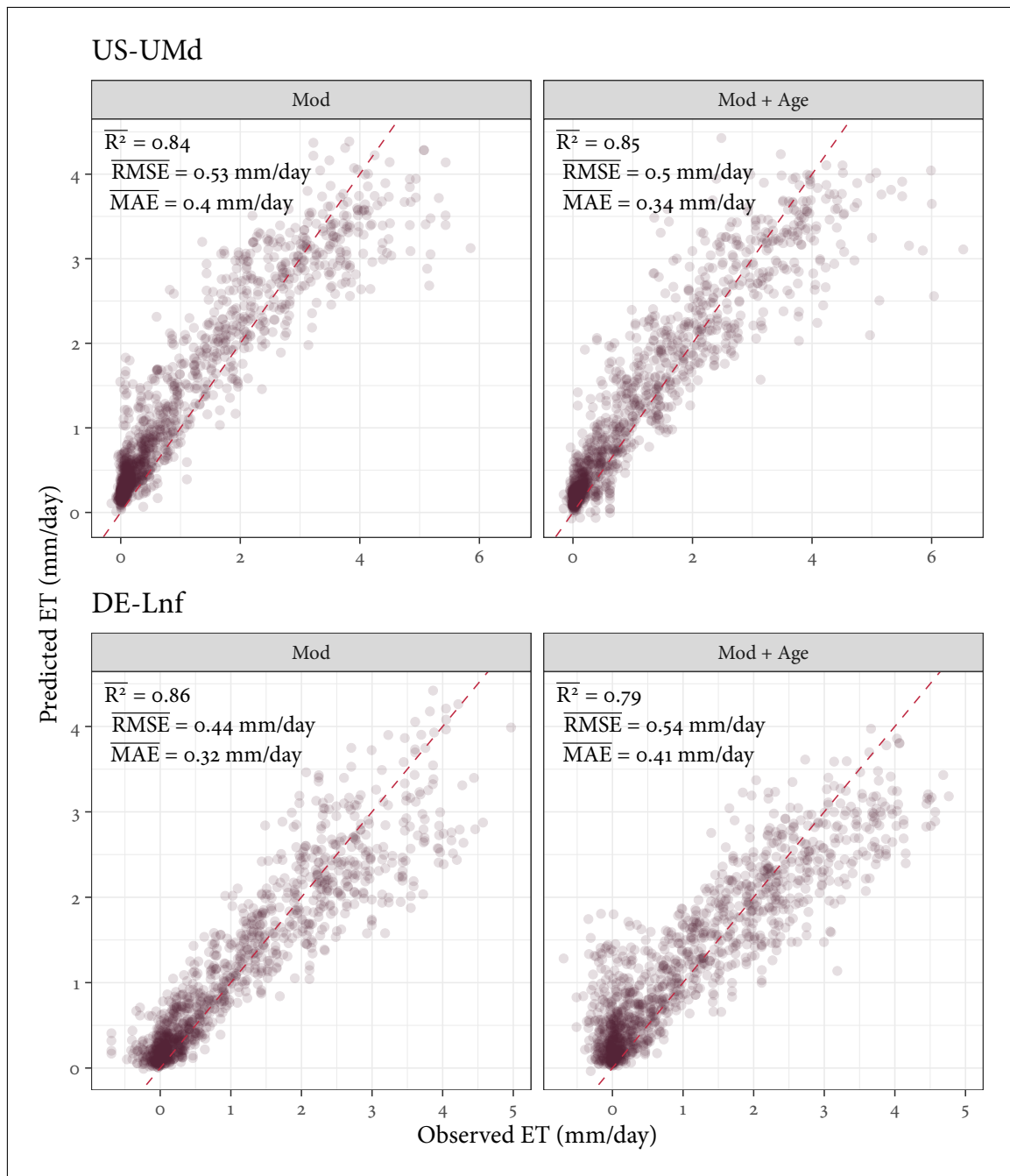


Figure 7: P

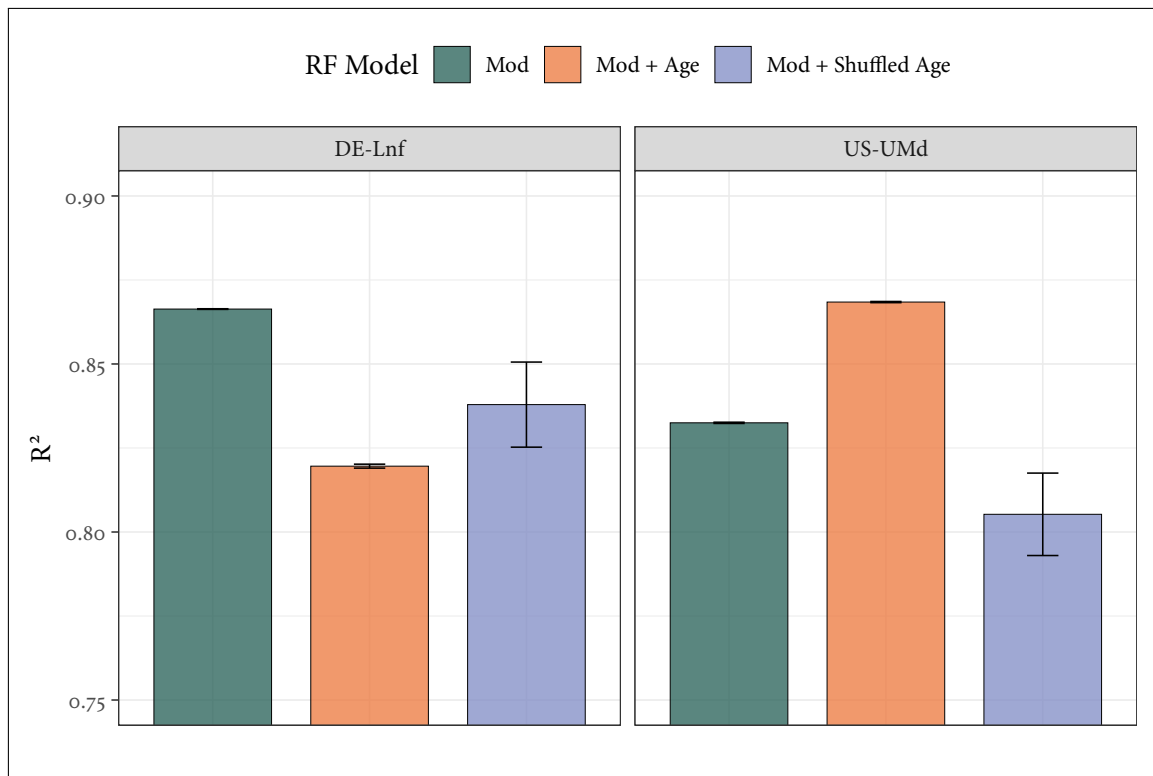


Figure 8: Bar plot showing the effect of different model setups on R^2 at DE-Lnf and US-UMd, with error bars indicating standard error. Models were trained using a leave one group out (LOGO) approach, with five repeats using different NumPy seeds. To assess whether site age was acting as a proxy for other site-level characteristics in these models, a model which randomly shuffled ages between sites was also formed using identical seeds and LOGO approach. This ‘shuffled age’ model appeared to increase R^2 at DE-Lnf compared to the Mod + Age model where an opposite effect was observed at US-UMd, suggesting that true age-ET relationships learned by the model transfer well to predicting ET at US-UMd, but not at DE-Lnf.

4 DISCUSSION

5 CONCLUSION

REFERENCES

- [1] Gilberto Pastorello, Carlo Trotta, Eleonora Canfora, Housen Chu, Danielle Christianson, You-Wei Cheah, Cristina Poindexter, Jiquan Chen, Abdelrahman Elbashandy, Marty Humphrey, et al. “The FLUXNET2015 dataset and the ONEFlux processing pipeline for eddy covariance data”. In: *Scientific Data* 7.1 (July 2020), p. 225. ISSN: 2052-4463. DOI: [10.1038/s41597-020-0534-3](https://doi.org/10.1038/s41597-020-0534-3). URL: <https://doi.org/10.1038/s41597-020-0534-3>.
- [2] R Core Team. *R: A Language and Environment for Statistical Computing*. R Foundation for Statistical Computing. Vienna, Austria, 2026. URL: <https://www.R-project.org/>.
- [3] Hadley Wickham, Mara Averick, Jennifer Bryan, Winston Chang, Lucy D’Agostino McGowan, Romain François, Garrett Golemund, Alex Hayes, Lionel Henry, Jim Hester, et al. “Welcome to the tidyverse”. In: *Journal of Open Source Software* 4.43 (2019), p. 1686. DOI: [10.21105/joss.01686](https://doi.org/10.21105/joss.01686).
- [4] Simon Besnard, Nuno Carvalhais, M Altaf Arain, Andrew Black, Sytze De Bruin, Nina Buchmann, Alessandro Cescatti, Jiquan Chen, Jan GPW Clevers, Ankur R Desai, et al. “Quantifying the effect of forest age in annual net forest carbon balance”. In: *Environmental Research Letters* 13.12 (2018), p. 124018.
- [5] R Myneni, Y Knyazikhin, and T Park. *MCD15A3H MODIS/Terra+ Aqua Leaf Area Index/FPAR 4-day L4 Global 500m SIN Grid V006, NASA EOSDIS Land Processes DAAC [data set]*. 2015.
- [6] Koen Hufkens. “bluegreen-labs/MODISTools: MODISTools v1. 1.4 (CRAN release)”. In: *Zenodo* (2023).
- [7] Sabine Landau and I Chis Ster. “Cluster analysis: overview”. In: *International encyclopedia of education* 3 (2010), pp. 72–83.
- [8] John C Gower. “A general coefficient of similarity and some of its properties”. In: *Biometrics* (1971), pp. 857–871.
- [9] Peter J Rousseeuw and L Kaufman. “Finding groups in data”. In: *Hoboken: Wiley Online Library* 1 (1990), p. 371.
- [10] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, et al. “Scikit-learn: Machine learning in Python”. In: *the Journal of machine Learning research* 12 (2011), pp. 2825–2830.
- [11] Takuya Akiba, Shotaro Sano, Toshihiko Yanase, Takeru Ohta, and Masanori Koyama. “Optuna: A next-generation hyperparameter optimization framework”. In: *Proceedings of the 25th ACM SIGKDD international conference on knowledge discovery & data mining*. 2019, pp. 2623–2631.

- [12] Mukund Palat Rao, Arturo Pacheco-Solana, Rong Li, Bar Oryan, Johanna E Jensen, Milagros Rodriguez-Caton, Lily Klinek, Zoe A Pierrat, Sophie Ruehr, Rose Oelkers, et al. “Decoupled carbon assimilation and growth responses to aridity in temperate deciduous oaks”. In: *Science advances* 12.24 (2026), eady7139.

6 FIGURES AND TABLES

[space for plot as in Rao et al. [12] (fig. 3, E/F)]

7 SUPPLEMENTARY MATERIALS

All scripts used in this study, alongside a guide of how to re-run the full analysis and package versions, can be found at: <https://github.com/archiebenn/forest-age-ET.git>